

StrataGrid™ uniaxial (SGU) is a polyester (PET) geogrid reinforcement for soil. These high-performance geogrids are constructed of high molecular weight polyester yarns knitted with a proprietary coating. SGU grid is engineered to be mechanically and chemically durable, in both the harsh construction installation phase and in the aggressive soil environments.

		Unit	SGU 500	SGU 600	SGU 700	SGU 800	SGU 900	SGU 1000
Mechanical properties								
Characteristic tensile strength (ASTM D6637 / EN ISO 10319) (min.)	MD	kN/m	500	600	700	800	900	1000
	CMD	kN/m	30	30	30	30	30	30
Elongation at characteristic strength ⁽¹⁾ (ASTM D6637 / EN ISO 10319)	MD	%	10.5	10.5	10.5	10.5	10.5	10.5
Tensile strength at ⁽²⁾ (ASTM D6637 / EN ISO 10319)	2% strain	MD	kN/m	75	90	105	120	135
	3% strain		kN/m	100	120	140	160	180
	5% strain		kN/m	200	240	280	320	300
Creep reduction factor (at 20°C, 120 years design life)	RF _{CR}		1.42	1.42	1.42	1.42	1.42	1.42
Creep limited strength	MD	kN/m	352.1	422.5	492.9	563.3	633.8	704.2
Partial factor – installation damage (ASTM D5818) (In clay, silt or sand)	RF _{ID}		1.06	1.06	1.06	1.06	1.06	1.06
Partial factor - environmental effect (5 ≤ pH ≤ 8)	RF _D		1.10	1.10	1.10	1.10	1.10	1.10
Physical properties								
Roll width		m	5.7	5.7	5.7	5	3.8	3.8
Roll length		m	100	100	100	100	100	100
Roll Weight (±5%)		kg	700	798	900	885	730	986
Molecular properties								
Molecular weight (ASTM D4603)		g/mol	min. 25,000					
Carboxyl end group (CEG) (ASTM D7409)		mmol/kg	max. 30					

Notes

- The above properties might change at the time of the manufacturing, storing, handling or shipping. The values are subject to change as per the discretion of the company.
- Varying roll widths can be provided on order - 1.9m / 3.8m / 5m.
- Customised rolls with varying lengths or master rolls can be manufactured on order.
- Tested in GAI-LAP and NABL accredited laboratories. ISO 9001:2015, ISO 14001:2015, ISO 45001:2018 Certified Organisation.

⁽¹⁾Tolerance for elongation is ±2.5%.

⁽²⁾Tensile strength in terms of characteristics (90th percentile) values, which are statistically safe values.

Reference: SGIPL-2526-TD-TDS-SGU-D5R2-02

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www.strataglobal.com | technical@strataindia.com | +91 22 4063 5100



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